MILITARY SPECIFICATION

RADIO SET AN/ARC-131( )
(Units of)

1. SCOPE

1.1 This specification covers one type of multi-channel frequency modulated airborne communication radio set which operates over the range of 30.00 to 75.95 MHz and is designated Radio Set AN/ARC-131( ) (See 6.1 & 6.3).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this document to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-636	Box, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple Wall
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-S-760	Strapping, Nonmetallic, (and Connectors)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-E-1	Electron Tube, General Specification for
MIL-P-116	Preservation, Methods of
MIL-T-713	Twine, Impregnated, Lacing and Typing (For Use in Electrical and Electronic Equipment)
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements Aircraft Equipment
MIL-M-10578	Metal Conditioner and Rust Remover (Phosphoric Acid Base)
MIL-M-13231	Marking of Electronic Items

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MIL-P-13949	Plastic Sheet, Laminated, Copper Clad (For Printed Wiring)
MIL-F-14072	Finish for Ground Signal Equipment
MIL-S-19500	Semiconductor Device, General Specification for
MIL-B-43291	Boxes, Fiberboard, Corrugated, Double Wall, Weather Resistant
MIL-I-46058	Insulating Compound, Electrical (For Coating Printed Circuits Assemblies)
MIL-P-55110	Printed Wiring Boards

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized Unit Loads (40 in. x 48 in. 4-Way Partial and 4-Way Pallets)
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-275	Printed Wiring for Electronic Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-704	Electrical Power, Aircraft, Characteristics and Utilization of
MIL-STD-810	Environmental Test Methods

DRAWINGS

USAECOM

SC-DL-619455	Receiver Transmitter, Radio RT-823( )/ARC-131
SC-DL-620155	Control, Radio Set C-7088( )/ARC-131
SC-DL-620305	Mounting MT-3664( )/ARC-131
SC- A-678005	Combined Burn-In/Reliability Test Procedure for Radio Set AN/ARC-131( )

(Copies of specifications, standards, and publications required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

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the two documents conflict, the ... shall apply unless

3. REQUIREMENTS

3.1 Description. - Radio Set AN/ARC-131 is a light weight multichannel airborne radio set capable of transmitting and receiving frequency modulated voice communications on anyone of 920 discreet channels spaced 50 kHz throughout the frequency range of 30.00 to 75.95 MHz. Frequency selection for the transmitter and receiver is remotely controlled and the selected frequency displayed on the operator's control box. Retransmission and homing facilities are provided. The Radio Set operates from an aircraft power supply having a nominal voltage of 27.5 volts DC.

3.2 Construction. - The equipment shall be constructed in accordance with the following applicable drawing and data lists:

SC-DL-619455	Receiver Transmitter, Radio	RT-823( )/ARC-131
SC-DL-620155	Control, Radio Set	C-7088( )/ARC-131
SC-DL-620305	Mounting	MT-3664( )/ARC-131

3.3 Preproduction Sample (See 4.3 & 6.2). - The contractor shall furnish three (3) preproduction samples for approval, if required by the invitation for bids and contract.

3.4 Printed Wiring Assemblies (See 4.4). - Printed wiring assemblies shall be processed for compliance with the requirements specified, in order not to affect life and service ability of the assemblies.

3.4.1 Printed Wiring Boards. - Printed wiring boards used in the construction of printed wiring assemblies, shall be a product which has passed the requirements of Specification MIL-P-55110, Standard MIL-STD-275 and the master drawing. However, those parts of Standard MIL-STD-275 which are in conflict with the applicable printed wiring board drawings and which require redesign of the printed wiring boards shall not apply. Conflicts between master drawings and Standard MIL-STD-275 need not be submitted as required by Specification MIL-P-55110. The printed wiring assemblies shall be clean and free from dirt, oil, corrosion, corrosion products, salt, smut, grease, fingerprints and all other foreign matter. There shall be no whiskers or solder or plating left on the conductor pattern. Printed wiring assemblies shall be processed in such a manner as to meet the requirements specified, to be uniform in quality and to be free from defects that will affect life and serviceability.

3.4.2 Laminate. - The laminate shall be type GB or GE in accordance with Specification MIL-P-13949, unless otherwise specified in the master drawing.

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3.4.3 Plating. - Plating of printed wiring boards shall be in accordance with Specification MIL-P-55110, and the master drawing.

3.4.4 Protective Coating. - Printed wiring assemblies shall be protected on both sides of the printed wiring assembly with an epoxy-type conformal coating to protect the pattern against moisture, dust and other contaminants. The coating shall be in accordance with Specification MIL-I-46058. The thickness of the coating shall be .001 IN. MIN, TO .008 IN. MAX. Assemblies having adjustable components shall not have the adjustable portion covered with the coating. Electrical and mechanical mating surfaces such as connector contact points, test points, screw threads, bearing surfaces, etc shall not be coated. Hermetic sealing of the assembly in a dry atmosphere shall be acceptable in place of conformal coating.

3.4.5 Solderability. - Solderability of printed wiring boards shall conform to Specification MIL-P-55110.

3.5 Finish, Protective (See 4.4). - Unit(s) shall be given protective finish in accordance with Specification MIL-F-14072 and the unit(s) drawings. This includes finish of hardware such as hinges, screws, etc., and necessary touch-up after mounting. The final paint film shall be Color No. 37038 (Lusterless Aviation Black) in accordance with MIL-F-14072.

3.6 Marking (See 4.4). -

3.6.1 General. - Marking shall conform to Specification MIL-M-13231. Front panel marking shall be group 1 as described in that specification.

3.6.2 Visibility. - Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the unit(s).

3.6.3 Serial Numbers. - All of the following unit(s) on order shall be serial numbered.

Control, Radio Set	C-7038( )/ARC-131
Mounting	MT-3564( )/ARC-131
Receiver Transmitter, Radio	RT-823( )/ARC-131

3.7 Cleaning. -

3.7.1 Parts. - After fabrication, parts shall be cleaned in accordance with requirement 9 of MIL-STD-454, or as specified in an applicable document.

3.7.2 Units. - After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

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3.8 Welding. - Welding shall be in accordance with the equipment drawing. Spot or other intermittent welds in aluminum or magnesium shall be limited to lightly stressed, noncritical structures. When spot or other intermittent welds are used to hold a part, the numbers of welds shall be at least two.

3.8.1 Cleaning Prior to Welding. - Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material.

3.8.2 Process. - Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, and other harmful defects. Where undesirable internal stresses are likely to result from welding, welded items shall be stress-relieved. Inert-gas-shielded arc welding shall be used, when practicable, for welding of aluminum or magnesium.

3.8.3 Cleaning After Welding. - Welded assemblies shall be cleaned to remove rust, scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Prior to painting, steel parts that have been arc welded or acetylene welded shall, in addition, be subjected to vat passivation or a phosphoric acid etch in accordance with Specification MIL-M 10578. Acid used for cleaning shall be completely neutralized and removed.

3.9 Riveting. - After riveting, the joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.10 Securing of Parts. -

3.10.1 General. - Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall retain items securely when subjected to specified service conditions.

3.10.2 Securing of Parts by Threaded Fasteners. - Nylok or other screws with plastic devices, and Loctite or similar sealants shall not be subjected to temperatures in excess of 250°F (as during baking of paint). Loctite or similar sealants shall be applied in accordance with the manufacturer's instructions.

3.11 Soldering. -

3.11.1 Solder. - Solder shall conform to the equipment drawings.

3.11.2 Acid or Acid Salts. - No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid salts be used where they can come in contact with insulation material.

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Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.11.3 Process. - There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.12 Wiring and Cabling. - Wiring and cabling shall be neat and sturdy.

3.12.1 Cabling. - Insulated wires shall be formed into cables except where operation of the equipment would be adversely affected thereby or where it is physically impracticable as in the case where the resulting cables would be excessively large and would interfere with operation or maintenance. The cabling or wires shall be effected by tape or twin conforming to Specification MIL-T-713. Individual conductors thus secured lie essentially parallel; however, this does not prohibit the use of twisted pairs.

3.12.2 Slack. - Wires and cables shall be as short as practical except that sufficient slack shall be provided for the following purposes:

a. To prevent undue stress on cable forms, wires, and connections, including connections to resiliently supported parts.

b. To enable parts to be removed and replaced during servicing without disconnecting other parts.

c. To provide for at least two replacements of the part to which the wire or cable is connected, except for certain radio-frequency leads which must be as short as possible for electrical reasons.

d. To facilitate field repair of broken or cut wires.

e. To prevent chafing or breaking of individual wires due to repeated flexing hinged parts.

3.12.3 Protection. - Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

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3.12.4 Support. - Wire and cable shall be properly supported and secured, to prevent undue stress on the conductors and terminals and undue change in position of the wire or cable (i) during and after subjection of the equipment to specified service conditions or (ii) after service or repair of the equipment in a normal manner.

3.12.5 Clearance. - Clearance between wires or cables, and parts such as electron tubes and resistors, shall be provided to avoid determination of the wires or cables because of the heat dissipated by such parts when the equipment is subjected to specified service conditions. Clearance between solder connections or bare conductors, on terminal boards, relays, or other parts, shall be such that no accidental contact can occur between adjacent connections when subjected to specified service conditions.

3.12.6 Splicing. - Wires in a continuous run between two terminals shall not be spliced during the wiring operation.

3.12.7 Connections. -

3.12.7.1 General. - Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength or solder alone. Where practicable, wires soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver plated, or lead-alloy coated. Electrical connections shall not be made by clamping between a metallic and non-metallic material.

3.13 Controls. - All movable controls shall operate properly without binding or other undue restriction. Controls shall not be assembled in a misaligned condition.

3.14 Tubes, Electrons and Semiconductors, (See 4.4). - These electron devices (tubes and semiconductors) shall conform to Specification MIL-E-1 and MIL-S-19500.

3.15 Tropicalization of Materials. - Materials shall be treated as follows:

3.15.1 Treating Materials.- Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating compounds shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.15.2 Toxicity. - Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

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3.15.3 Flexibility. - Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.15.4 Statement of Treatment. - The statement of treatment shall indicate whether or not nutrient materials have been employed. If nutrient materials have been used, it shall describe the materials in detail and the treating compounds and processes that the contractor proposes to use. (See 6.2)

3.16 Service Conditions (See 4.8). - The units shall meet the following service conditions as Class 1 equipment as specified in Specification MIL-E-5400 and meet the performance specification, except as otherwise specified herein.

3.16.1 Temperature (See 4.8.1). -

a. Operating:- Temperature-Altitude per MIL-E-5400 Class 1 equipment except Curve B limited to +65°C.

b. Non-Operating:- From -54°C to +71°C.

3.16.2 Altitude (See 4.8.2). - Non-Operating and

Operating:- Temperature-Altitude per MIL-E-5400 Class 1 equipment except Curve B limited to +65°C.

3.16.3 Relative Humidity (See 4.8.3). - Up to 100 percent relative humidity including conditions wherein condensation takes place in the form of water. This shall be performed over a 10 day cycle.

3.16.4 Orientation, Operating, and Non-Operating (See 4.8.4). - Indefinite storage in any orientation.

3.16.5 Sand and Dust, Operating and Non-Operating (See 4.8.5). - The units shall withstand exposure to sand and dust particles as encountered in desert areas and meet the tests as specified.

3.16.6 Vibration (See 4.8.6). - The units shall meet the vibration requirements and performance indicated.

3.16.7 Shock (See 4.8.7). -

3.16.7.1 Bench Handling (See 4.8.7.1). - The units shall meet required performance after the shock test.

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3.16.7.2 Equipment Shock (See 4.8.7.2). - The units shall meet the equipment shock test. The units shall meet performance specified after the shock test.

3.16.8 Mounting (Crash Safety) (See 4.8.8). - The mounting shall meet the crash safety test of MIL-E-5400. Bending and distortion shall be permitted, but there shall be no failure of the attaching joints, and the equipment or dummy load shall remain in place.

3.16.9 Explosion Proof (See 4.8.9). - The units shall be explosion proof per Specification MIL-E-5400.

3.16.10 Fungus (See 4.8.10). - The unit(s) shall show no evidence of viable fungus or corrosion after being subjected to 28 days of fungus test.

3.16.11 Salt Fog (See 4.8.11). - The unit(s) shall be exposed, in normal operating configuration, to salt fog for 48 hours and shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered or otherwise loosened finish.

3.17 Transmitter - (See 4.9).

3.17.1 Power Output, RF (See 4.9.1). - With the transmitter operating into a 52 OHM resistive load a minimum of 10 watts and no more than 18 watts of RF Power shall be delivered to the load for any variation of service conditions with the following exception: At service conditions from -40°C to -54°C, the transmitter power output shall be a minimum of 10 watts from 30 to 70 MHz and shall be a minimum of 7 watts from 70.05 to 75.95 MHz. The XMIT HI-LO switch shall reduce the power output to 1 watt ± 3 db when in the LO position.

3.17.2 Distortion. - (See 4.9.2).

3.17.2.1 Distortion, Narrow Band (See 4.9.2.1). - The overall harmonic distortion shall not exceed 5 percent.

3.17.2.2 Distortion, Wide Band (See 4.9.2.2). - The overall total harmonic distortion shall not exceed 7 percent for all modulating frequencies up to 20,000 Hz.

3.17.3 Sidetone (See 4.9.3). - The sidetone output shall be furnished through the audio section of the receiver and the presence of a sidetone signal shall be an indication that radio frequency power is present at the antenna terminals of the transmitter. When the carrier is modulated, at the standard 8 kHz deviation, the sidetone output shall be 6 ± 2 db below the receiver output. Total harmonic distortion of the sidetone output shall be 7 percent or less. Sidetone fidelity shall be within + 3 to - 5 db of the level at 1000 Hz for any frequency from 500 to 3000 Hz.

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3.17.4 Modulation Capability (See 4.9.4). - The transmitter shall be capable of being deviated in accordance with the following, without breaking lock.

Frequency (Hz)	Deviation (kHz) Minimum
500	25
1000	25
2000	25
3000	25
10000	25
20000	25

3.17.5 Modulator Sensitivity (See 4.9.5). - The modulator sensitivity shall be in accordance with the following:

BAND	TEMPERATURE	MODULATOR SENSITIVITY kHz/V
A	77°F	240 to 650
A	-40°F and 150°F	230 to 650
B	77°F	275 to 650
B	-40°F and 150°F	265 to 665

3.17.6 Resetability (See 4.9.6). - The total backlash to the transmitter VFO shall not exceed 100 kHz.

3.17.7 Frequency Accuracy (See 4.9.7). - The frequency accuracy of the transmitter shall be within ± 3.5 kHz of the dial frequency.

3.17.8 Maximum Signal-Plus Noise to Noise Ratio (See 4.9.8). - When measured in accordance with 4.9.8, the maximum overall signal-plus noise to noise ratio of the transmitter shall not be less than 35 db.

3.17.9 Audio Response (See 4.9.9). -

3.17.9.1 Narrow Band Audio Response (See 4.9.9.1). - The frequency response between 500 to 3000 Hz shall be within ± 2 db of a 1000 Hz reference, and shall be down a minimum of 14 db at 5000 Hz and above.

3.17.9.2 Wide Band Audio Response (See 4.9.9.2). - The frequency response from 50 to 20,000 Hz shall be within ± 2 db of a 1000 Hz reference.

3.17.10 Audio Input Circuit Impedance (See 4.9.10). - The nominal impedance of this circuit shall be 150 OHMS ± 10 percent at frequencies in the range of 500 to 3000 Hz.

3.17.11 Spurious Radiations (See 4.9.11). - When measured as specified the number of spurious outputs which exceed 10 dB of quieting shall not be greater than 6 for each test frequency.

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3.17.12 Deviation (See 4.9.12). - The nominal deviation of the transmitter shall be 8 ± 2 kHz.

3.17.13 VFO Accuracy (See 4.9.13). - The VFO frequency shall be within 350 kHz of the dial frequency between 30.00 and 52.95 MHz and 450 kHz of the dial frequency between 53.00 and 75.95 MHz. (These limits include the 100 kHz allowed for resetting.) The accuracy shall be within 250 kHz between 30.00 and 52.95 MHz and 350 kHz between 53.00 MHz and 75.95 MHz if the frequencies are not reset.

3.17.14 Tone Deviation (See 4.9.14). - Operation of the tone of 150 Hz ± 2 Hz shall deviate the transmitter 3 kHz ± 500 Hz. This tone shall be transmitted at all times when the transmitter is keyed. The tone transmission, when using an external security device, shall be disabled.

3.17.15 Catching Range (See 4.9.15). - The catching range of the transmitter shall be more than 750 kHz in both plus and minus directions.

3.17.16 Continuous Transmit Operation (See 4.9.16). - The transmitter shall be capable of three-hour continuous transmission at the high power output level without failure or degradation of performance below established specification limits.

3.18 Receiver (See 4.10). -

3.18.1 Sensitivity (See 4.10.1). - RF signal inputs of 0.5 microvolts at room ambient temperature shall produce a signal-plus-noise-plus distortion to noise-plus distortion ratio of 10 db minimum. At -54°C and + 65°C an RF input of 0.6 microvolts shall produce a S+N+D/N+D ratio of 10 db minimum.

3.18.2 Squelch Sensitivity (See 4.10.2). -

3.18.2.1 Tone Operation (See 4.10.2.1). - The tone operated squelch shall open with an RF signal of 0.5 microvolt or less. At the temperature extremes, (-54°C and +65°C), the tone squelch shall operate at 0.8 microvolt or less. Faulty operation of the squelch shall not occur when the incoming signal is voice modulated. The squelch, once having opened, shall remain open when the RF signal is reduced to 0.4 microvolt. At the temperature extremes (-54°C and +65°C), the squelch once having opened, shall remain open when the RF signal is reduced to 0.7 microvolt. The difference between the pull-in and drop-out signal levels shall be less than 6 db.

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3.18.2.2 Noise Operation (See 4.10.2.2). - The noise operated squelch shall open with an RF signal of 0.7 microvolt or less. At the temperature extremes (-54°C and +65°C), the noise squelch shall operate at 0.8 microvolt or less. Faulty operation of the squelch shall not occur when the incoming signal is voice modulated. The squelch once having opened, shall remain open when the RF signal is reduced to 0.55 microvolt. At the temperature extremes (-54°C and +65°C), the squelch once having opened, shall remain open when the RF signal is reduced to 0.7 microvolt. The difference between the pull-in and drop-out signal levels shall be less than 6 db.

3.18.2.3 Maximum Squelch Level (See 4.10.2.3). - When the squelch adjust control is in a full clockwise position, the signal required to unsquelch the receiver shall be no more than 1.5 microvolts. (CARRIER SQUELCH MODE)

3.18.2.4 Squelch Release (See 4.10.2.4). - An instantaneous application of a signal level of 1.0 volt or more at the receiver RF input, shall not cause a delay in the release of the squelch of more than 250 milliseconds.

3.18.3 IF Selectivity (See 4.10.3). - The IF selectivity shall be as follows:

a. Band width 6 db points, 32 kHz minimum

b. Band width 60 db points, 80 kHz maximum

c. Band width 85 db points, 100 kHz maximum

The response characteristics in the passband shall have no irregularities greater than 2.5 db.

3.18.4 Maximum Signal-Plus-Noise to Noise Ratio (See 4.10.4). - The maximum signal-plus-noise to noise ratio shall not be less than 45 db.

3.18.5 Desensitization (See 4.10.5). - When measured in accordance with 4.10.5, a 25 millivolts interfering signal which is ± 500 kHz removed from the desired signal shall not degrade a 26 db signal-plus-noise to noise signal by more than 6 db.

3.18.6 Audio Output (See 4.10.6). - The receiver audio output termination shall be 150 OHMS ungrounded.

3.18.6.1 Narrow Band Audio Output (See 4.10.6.1). - The receiver audio output shall be 50 MW ± 2 db when measured across a 150 OHM resistive load.

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3.18.6.2 Wide Band Audio Output (See 4.10.6.2). - The wideband circuit shall deliver 1.4 to 2.5 volts into a 50,000 OHM load when receiving a signal having 8 kHz deviation.

3.18.7 Audio Response (See 4.10.7). -

3.18.7.1 Narrow Band Audio Response (See 4.10.7.1). - The overall audio response of the receiver shall be within ± 2 db of 1000 db reference from 500 to 3000 Hz. At 150 Hz the response shall be down at least 10 db, and at 6 kHz and above, the response shall be down at least 20 db.

3.18.7.2 Wide Band Audio Response (See 4.10.7.2). - The overall audio response of the receiver when measured at the wideband output shall be within + 2 db, - 3 db of the 1000 Hz reference from 50 to 12,000 Hz.

3.18.8 Limiting (See 4.10.8). - The limiting characteristics of the receiver shall be such that the audio output shall not vary by more than ± 1 db.

3.18.9 Distortion (See 4.10.9). -

3.18.9.1 Narrow Band Distortion (See 4.10.9.1). - The total harmonic distortion shall be less than 8 percent. This requirement shall be met from 500 to 3000 Kz and carrier deviation of ± 8 kHz.

3.18.9.2 Wide Band Distortion (See 4.10.9.2). - Overall wideband harmonic distortion shall not exceed 5 percent with the narrow band IF filter removed.

3.18.10 Spurious Responses (See 4.10.10). - There shall be no more than 5 spurious responses beyond ± 500 kHz of the desired frequency which are less than 100 db down from a one microvolt reference, and no more than 5 spurious responses within ± 500 kHz of the desired frequency which are less than 80 db down from a one microvolt reference.

3.18.11 IF Rejection (See 4.10.11). - IF rejection shall not be less than 100 db.

3.18.12 Intermodulation (See 4.10.12). - Any intermodulation resulting from two undesired carrier frequencies of the same level shall be at least 58 db down from the desired response.

3.18.13 Oscillator Radiation (See 4.10.13). - The radiation of any oscillator within the equipment shall produce an output no greater than 75 microvolts when the receiver is tuned to any frequency within its tuning range.

3.18.14 V.F.O. (See 4.10.14). -

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3.18.14.1 Modulation Sensitivity (See 4.10.14.1). - The deviation sensitivity of the oscillator shall be within the limits of 300 kHz/V and 600 kHz/V maximum.

3.18.14.2 Deviation Capability (See 4.10.14.2). - The deviation of the oscillator shall not be less than 450 kHz nor greater than 950 kHz over the tuning range of the oscillator.

3.18.14.3 Frequency Accuracy vs Temperature (See 4.10.14.3). - The local oscillator frequency shall not differ from the dial frequency more than ± 180 kHz, including the 85 kHz allowed for resetting.

3.18.14.4 Resetability (See 4.10.14.4). - When measured as specified the dial resetability shall be such that the maximum frequency difference between any two of five settings to any point on the dial from either direction shall not be greater than 85 kHz.

3.18.15 Maximum RF Input to Receiver (See 4.10.15). - Regardless of the settings of the receiver controls, no damage shall result to any receiver unit from the operation of its companion transmitter or from the presence of an RF signal of 2.0 volts RMS into the receiver antenna terminals.

3.18.16 Catching Range (See 4.10.16). - The sum of the plus and minus catching ranges shall not be less than 850 kHz nor shall either the plus or minus be less than 250 kHz.

3.18.17 Residual Phase Detector Output (See 4.10.17). - The residual phase detector output voltage shall be 0.17 VDC or less.

3.18.18 Blocking (See 4.10.18). - When a signal having an amplitude of one volt RMS and a frequency equal to that to which the receiver is tuned is applied to the receiver terminals, it shall be possible to obtain a 10 db signal plus noise-to-noise ratio with a 1.0 microvolt input within 250 milliseconds following the removal of the one volt signal. No blocking shall occur in the receiver under push-to-talk operation.

3.19 Homing (See 4.11). -

3.19.1 Homing Sensitivity (See 4.11.1). - The sensitivity control of the homer module shall be adjusted such that an input of 4.8 microvolts shall provide sufficient input to the pads to indicate a flag down condition. The above measurements shall be made when equal energies are being received by each homing input, indicating an "on course" direction of the aircraft.

3.19.2 Visual Homing Indication (See 4.11.2). - The homing system shall operate so that movements of the needle and flags shall indicate the following conditions:

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a. With the vertical centering control properly adjusted, equal signal levels through two 6.0 DB pads of not more than 4.8 microvolts (input to the pads) shall result in a flag down condition and the vertical needle being in center position (±10UA).

b. With 12 microvolts input to the pads and the left pad adjusted to 3.0 DB (Right pad 6.0 DB) the vertical needle shall deflect to the left. Right deflection shall exist when the left pad is adjusted to 3.0 DB. The right deflection shall be within 20 percent of the left deflection.

c. With signal levels from 12 to 12,000 microvolts input to the pads and 3.0 DB differential between the pads, the needle deflection left or right shall not deflect less than the wing span on the horizontal scale.

3.19.3 Isolation of Communication Antenna Input During Homing (See 4.11.3). - The homing left-right indication displayed shall not be degraded more than 10% when an RF signal up to 5 millivolts is applied to the communications antenna input simultaneously with application of a 20 DB weaker RF signal to one homer antenna input and a 23 DB weaker RF signal (3 DB differential on homing) to the other homer antenna input.

3.20 Frequency Selection (See 4.12). - Frequency selection shall be actuated by Control C-7088/ARC-131 and shall tune the receiver-transmitter automatically to any channel within seven (7) seconds under the longest possible tuning cycle. During channeling the audio output shall consist of a channel tone of approximately 800 HZ.

3.21 Input Voltage (See 4.13). -

3.21.1 Input Voltage Range (See 4.13.1). - The RF unit shall meet the requirements specified when operated at any input voltage from 24.0 to 29.0 VDC.

3.21.2 Transient Protection (See 4.13.2). - The Radio Set AN/ARC-131 shall be protected against transients as specified in the MIL-STD-704, Limit 1.

3.22 Retransmission (See 4.14). - The Radio Set AN/ARC-131 shall be capable of retransmitting and be such as to provide the pilot with capabilities to monitoring either radio channel during retransmission.

3.23 Radio Interference (See 4.15). - The units shall meet the requirements of MIL-I-6181D except as follows:

3.23.1 Conducted Interference. - This test to be performed on the power line only. The values indicated in Figures 7, 8, 9 and 10 shall be exceeded by not more than 15 DB (Figures Reference MIL-I-6181).

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3.23.2 Radio Frequency Conducted (Para 4.3.4.1.1 MIL-I-6181). - This test shall be conducted on power lines only. The frequency range shall be 0.150 MHz to 30,000 MHz. The signal-to-noise ratio of the receiver shall not be degraded by more than 2 db.

3.23.3 Transmitter Kay-up or Receiver (Para 4.3.3.1 MIL-I-6181). - The RF output of any transmitter key-up or receiver shall not exceed 80 db above a one microvolt for CW operation.

3.23.4 Broadband Intermodulation (Para 4.3.4.4.2 MIL-I-6181). - This test shall not apply.

3.23.5 Two Signal Intermodulation (Para 4.3.4.4.1 MIL-I-6181D). - The magnitude of each signal generator shall be 70 db above one microvolt.

3.23.6 Audio Frequency Conducted (Para 4.3.4.1.2 MIL-I-6181). - This test shall be conducted on power lines only. With one volt RMS open circuit applied as in Figure 19 of MIL-I-6181D, the following shall occur: With the squelch control set for 1.2 microvolts at the operating frequency and with no signal input, the audio output shall be noted. When audio frequencies of 50 to 15,000 Hz are applied as in Figure 19, the audio output of the receiver shall be at least 40 db below normal output.

3.23.7 Radiated Interference. - The radiated interference shall not exceed the limits established by paragraph 4.3.2 of MIL-I-6181 , by more than 10 db. This test shall not be conducted on carrier frequency.

3.23.8 Susceptibility (Para 4.3.4.2 MIL-I-6181). - Radio Frequency Radiated - Not more than 3 db of receiver quieting shall be produced when the equipment is subjected to the radio frequency field of this test.

3.24 X-Mode (See 4.16). - The equipment shall be compatible and operate satisfactorily when connected to Keyer KY-28 or KY-8.

3.25 Interchangeability (See 4.17). - Like units, assemblies, subassemblies, modules, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. Individual items shall not be handpicked for fit or performance; however, matched parts or set, when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.26 Burn-In/Reliability (See 4.6). - Each equipment on order shall be capable of meeting the requirements herein, without subsequent processing, after subjection to and meeting the burn-in/reliability requirement.

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3.27 Operational Inspection (See 4.18). - Each unit on order shall be subjected to the operation inspection as specified, after applicable processes, inspection and tests are completed, to determine that the meter on the front panel indicates +24 volts, +16.0 volts, AGC Voltage (homing), RF, FWD, RF Refl and OFF, at the applicable position of the meter switch, and that each unit is operable prior to packaging.

3.28 Technical Literature. - Technical literature shall be furnished as specified in the contract. Parts shall be identical to corresponding parts in the equipment furnished on the order (See 6.2).

3.29 Workmanship. - The units shall be manufactured and assembled in accordance with the equipment drawings, Requirement 9 of MIL-STD-454 and the following paragraphs of this specification:

Soldering	3.11
Wiring and Cabling	3.12
Cleaning	3.7
Welding	3.8
Riveting	3.9
Securing of Parts	3.10
Controls	3.13
Tropicalization	3.15

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.2 Classification of Inspection. - Inspection shall be classified as follows:

a. Preproduction inspection (does not include preparation for delivery). (See 4.3)

b. Inspection covered by subsidiary documents (See 4.4).

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

four edges and at 1-inch intervals between adjacent figures in the center.

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(2) Quality conformance inspection of preparation for delivery. (See 4.20)

4.3 Preproduction Inspection. - This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for Group "A", Group "B", and Group "C" (See Tables I, II, and III, respectively).

The preproduction inspection will normally be performed in this order: (1) vibration, (2) shock, benchhandling and (3) shock, equipment. Other preproduction inspection may precede, follow, or be interspersed between the foregoing.

4.4 Inspection Covered by Subsidiary Documents. - The following shall be inspected under the applicalbe subsidiary documents as part of the inspection of equipment before preparation for delivery:

ITEM	WHERE REQUIRED
Printed Wiring Assemblies	3.4
Finish Protective	3.5
Marking	3.6
Electron Tubes & Semiconductors	3.14

4.5 Quality Conformance Inspection of Equipment Before Preparation For Delivery. - The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government - at its discretion - may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.6)

Test equipment for Government verification inspection shall be made available by the contractor. All Radio Sets AN/ARC-131( ) on order shall be subjected to the Burn-In/Reliability after final assembly. (See 3.26)

4.5.1 Group "A" Inspection. - Each unit of each lot of equipment shall be inspected for conformance to all the inspection and test requirements of Table I. Each lot will be subjected to verification, utilizing the procedures of MIL-STD-105, using the general inspection levels and the AGL's indicated in Table I. Group "A" inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection 4.18 shall be last (See 6.4).

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TABLE I - Group "A" Inspection

Inspection	Rqt Par.	Insp Par.	AQL Major	AQL Minor
Visual and Mechanical	3.29	4.19	-----	-----
Receiver Section			-----	-----
VHF Tuner boards:			-----	-----
Antenna board			0.65%	2.5%
1st RF board			0.65%	2.5%
2nd RF board			0.65%	2.5%
Mixer Board			0.65%	2.5%
Oscillator board			0.65%	2.5%
Tuner Power Supply board			0.65%	2.5%
VHF Tuner assembly:			1.0 %	4.0%
RF Oscillator assembly			1.0 %	4.0%
CRS Modules:			-----	-----
CRS Harmonic generator			0.65%	2.5%
CRS Balanced mixer			0.65%	2.5%
CRS 2nd mixer			0.65%	2.5%
CRS 1st and 2nd IF amplifier			0.65%	2.5%
CRS 3rd IF amplifier and limiter			0.65%	2.5%
CRS Hunt discriminator			0.65%	2.5%
CRS Phase discriminator			0.65%	2.5%
CRS Assembly			1.0 %	4.0%
IF Amplifier Modules:			-----	-----
1st and 2nd IF amplifier			0.65%	2.5%
3rd, 4th & 5th IF Limiter and discrimitor			0.65%	2.5%
Audio and Squelch preamplifier			0.65%	2.5%
IF Amplifier assembly			1.0 %	4.0%
Audio and Squelch Modules:			-----	-----
Audio amplifier			0.65%	2.5%
Squelch amplifier			0.65%	2.5%
Squelch filter			0.65%	2.5%
Audio and Squelch amplifier assembly			1.0%	4.0%
Transmitter Section			-----	-----
Oscillator-buffer boards:			-----	-----
Oscillator			0.65%	2.5%
Buffer			0.65%	2.5%
Oscillator-buffer assembly			1.0 %	4.0%
Voltage regulator			0.65%	2.5%
Homer module			0.65%	2.5%

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TABLE I - Group A Inspection (Cont'd)

Inspection	Rqt Par	Insp Par.	AQL Major	AQL Minor
RF control module			0.65%	2.5%
RF filter			0.65%	2.5%
RF attenuator and antenna switch			0.65%	2.5%
Reflectometer and sidetone detector			0.65%	2.5%
Isolation amplifier			0.65%	2.5%
Power Supply			0.65%	2.5%
RF amplifier			0.65%	2.5%
IF attenuator			0.65%	2.5%
Modulator boards:			0.65%	2.5%
XMTR 11.5 modulator			0.65%	2.5%
XMTR phase discriminator			0.65%	2.5%
XMTR 1st and 2nd IF amplifier			0.65%	2.5%
XMTR hunt generator			0.65%	2.5%
Audio input			1.0 %	4.0%
Modular Assembly			4.0 %	15.0%
Receiver Transmitter RT-823( )/ARC-131			1.0 %	6.5%
Control, Radio Set C-7088( )/ARC-131			1.0 %	4.0%
Mounting MT-3664( )/ARC-131				
Electrical				
Transmitter				
Power output	3.17.1	4.9.1		
Distortion, narrow band	3.17.2.1	4.9.2.1		
Sidetone	3.17.3	4.9.3		
Deviation	3.17.12	4.9.12		
Tone deviation	3.17.14	4.9.14		
Frequency accuracy (at rm temp only)	3.17.7	4.9.7		
Receiver				
Sensitivity (at rm temp only)	3.18.1	4.10.1	*1.0 % for the combined group	
Squelch sensitivity:				
Tone operation (at rm temp only)	3.18.2.1	4.10.2.1		
Noise operation (at rm temp only)	3.18.2.2	4.10.2.2		
Maximum squelch level	3.18.2.3	4.10.2.3		
Narrow band audio output	3.18.6.1	4.10.6.1		
Narrow band distortion	3.18.9.1	4.10.9.1		
Homing Sensitivity	3.19.1	4.11.1		
Visual homing indication	3.19.2	4.11.2		
Frequency selection	3.20	4.12		

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TABLE I - Group A Inspection (Cont'd)

Inspection	Rqt Par	Insp Par.	AQL Major	AQL Minor
Operational Inspection	3.27	4.18	0.1%	

* All electrical defects are considered major.

4.5.2 Group B Inspection. - This inspection, including sampling, shall conform to Table II and to the procedures for small-sample inspection of Standard MIL-STD-105 using the special inspection levels. The AQL shall be 6.5 percent defective and the inspection level shall be S4 for normal, tightened and reduced inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have met the Group A inspection.

TABLE II - Group B Inspection

Inspection	Rqt Par.	Insp Par.	AQL
Transmitter, Electrical:			
Modulation capability	3.17.4	4.9.4	
Resetability	3.17.6	4.9.6	
Maximum signal-plus-noise to noise ratio	3.17.8	4.9.8	
Narrow band, audio response	3.17.9.1	4.9.9.1	
Wide band, audio response	3.17.9.2	4.9.9.2	
Distortion, wideband	3.17.2.2	4.9.2.2	
Receiver, Electrical:			6.5% for the combined group
Wide band, audio output	3.18.6.2	4.10.6.2	
Maximum signal-plus-noise to noise ratio	3.18.4	4.10.4	
Narrow band, audio response	3.18.7.1	4.10.7.1	
Wide band, audio response	3.18.7.2	4.10.7.2	
Modulation Sensitivity	3.18.14.1	4.10.14.1	
Squelch Release	3.18.2.4	4.10.2.4	

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TABLE II - Group B Inspection (Cont'd)

Inspection	Rqt Par	Insp Par.	AQL
Deviation Capability	3.18.14.2	4.10.14.2	
Residual phase detector output	3.18.17	4.10.17	
Resetability	3.18.14.4	4.10.14.4	
IF selectivity	3.18.3	4.10.3	
Limiting	3.18.8	4.10.8	
Input Voltage	3.21.1	4.13.1	6.5%
Interchangeability (mechanical)	3.25	4.17	6.5%
Interchangeability (module performance)	3.25	4.17	6.5%

4.5.2.1 Order of Inspection Within Group B. - Group B inspection shall be performed in any order which is satisfactory to the Government.

4.5.3 Group C Inspection. - This inspection shall be as listed in Table III and shall normally be performed on sample units that have been subjected to and met Group A and Group B inspection. No defects shall be allowed. Sample unit(s) shall be selected in accordance with 4.5.3.1.

TABLE III - Group C Inspection

Inspection	Rqt Para.	Insp Para.
Subgroup I		
Transmitter, electrical		
Spurious radiation	3.17.11	4.9.11
VFO accuracy	3.17.13	4.9.13
Catching range	3.17.15	4.9.15
Frequency Accuracy (temp only)	3.17.7	4.9.7
Receiver, electrical		
Sensitivity	3.18.1	4.10.1
Squelch Sensitivity		
Tone operation	3.18.2.1	4.10.2.1
Noise operation	3.18.2.2	4.10.2.2
Spurious response	3.18.10	4.10.10
IF rejection	3.18.11	4.10.11
Intermodulation	3.18.12	4.10.12

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TABLE III - Group C Inspection (Cont'd)

Inspection	Rqt Para.	Insp Para.
Oscillator radiation	3.18.13	4.10.13
Frequency accuracy vs temperature	3.18.14.3	4.10.14.3
Catching range	3.18.16	4.10.16
Maximum RF input to receiver	3.18.15	4.10.15
Wideband distortion	3.18.9.2	4.10.9.2
Retransmission	3.22	4.14
Subgroup II		
Transmitter, electrical		
Audio input circuit impedance	3.17.10	4.9.10
Modulator sensitivity	3.17.5	4.9.5
Receiver, electrical		
Desensitization	3.18.5	4.10.5
Blocking	3.18.18	4.10.18
Subgroup III		
Orientation	3.16.4	4.8.4
Sand and dust	3.16.5	4.8.5
Bench handling	3.16.7.1	4.8.7.1
Equipment, shock	3.16.7.2	4.8.7.2
Mounting, (crash safety)	3.16.8	4.8.8
Explosion proof	3.16.9	4.8.9
Continuous Transmit Operation	3.17.16	4.9.16
Isolation, Communication Antenna during Homing	3.19.3	4.11.3
Transient Protection	3.21.2	4.13.2
Radio Interference	3.23	4.15
Fungus	3.16.10	4.8.10
Salt Fog	3.16.11	4.8.11
X-Mode	3.24	4.16
Subgroup IV		
Temperature	3.16.1	4.8.1
Altitude	3.16.2	4.8.2
Relative humidity	3.16.3	4.8.3
Vibration	3.16.6	4.8.6

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4.5.3.1 Sampling for Inspection of Equipment. - Units selected for each Group "C" inspection shall be selected without regard to their quality in accordance with the following:

4.5.3.1.1 Subgroup I. - For this subgroup, two units from the first 50 production units shall be selected for the first Group "C" inspection. For subsequent Group "C" inspections, two units from each successive 100, or two per month if less than 100 are prosuced monthly, shall be selected.

4.5.3.1.2 Subgroup II. - For this subgroup, two units from the first 50 production units shall be selected for the first Group "C" inspection. For subsequent Group "C" inspections, two units from each successive 250, or two every three months if less than 250 are produced during three successive months, shall be selected.

4.5.3.1.3 Subgroup III. - For this subgroup, two units from the first 50 production units shall be selected for the first Group "C" inspection; subsequent tests shall be conducted one time during the course of the production contract.

4.5.3.1.4 Subgroup IV. - For this subgroup, two units per month shall be selected.

4.5.3.2 Noncompliance. - The contractor shall immediately report in writing each Group "C" failure occurrance, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and, (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group "C" inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Groups "A" and "B" inspection may be reinstituted; however, final acceptance and shipment will be withheld until Group "C" reinspection results have shown that the corrective action was effective.

4.5.4 Reinspection of Conforming Group "C" Sample Units. - Unless otherwise specified, sample units which have been subjected to and passed Group "C" inspection may be accepted on the contract or order provided all damage is repaired and the sample units are resubjected to and pass Group "A" inspection.

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4.6 Combined Burn-In/Reliability Test. - Each unit on order shall be subjected to the burn-in reliability test, prior to quality conformance testing required by this specification, in accordance with SC-A-678005.

4.7 Standard Test Conditions. - Unless otherwise specified herein, all inspections shall be performed at room temperature, pressure and humidity under the following standard conditions:

a. Power Supply Voltage - 27.5 Volts DC ± 0.5 Volts.

b. Intermittent Duty Cycle - One minute transmit, 9 minutes receive.

c. Warm-up Time - Not more than 5 minutes.

d. Output Impedance - RF 50 OHMS, audio 150 OHMS.

e. Output Level - RF 10 watts, audio 50 MW.

f. Input Impedance - RF 50 OHMS, audio 150 OHMS

g. Input Level - Audio 0 DBM, at 1000 Hz.

h. Standard Test Signal - The standard test signal shall be a frequency modulated carrier deviated ± 8 kHz at a 1000 Hz rate produced by a signal generator having an internal impedance of 50 OHMS.

4.8 Service Conditions Tests (See 3.16). - The service conditions tests shall be conducted in accordance with MIL-T-5422 for Class 1 equipments; as defined in paragraphs of 3.16. Electrical tests shall be run during the operating time specified in each service condition test to determine compliance with paragraphs 3.17.1, 3.17.8, 3.17.12, 3.17.13, 3.18.1, 3.18.2.1, 3.18.2.2, 3.18.16 and 3.20 at supply voltages of 27.5 volts and carrier frequencies of 30.00, 50.00 and 75.00 MHz. The combined temperature altitude tests shall require the input voltage to be 29 VDC. A visual and mechanical inspection shall be performed after the humidity, vibration, salt fog, fungus and shock tests, no defects will be allowed.

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4.8.1 Temperature (See 3.16.1). -

a. Non-operating - From -54°C to +71°C.

b. Operating - Class 1 per MIL-T-5422 except Curve B limited to +65°C.

4.8.2 Altitude (See 3.16.2). - Sea level to 50,000 feet above sea level as defined per Class I MIL-T-5422 Curve B.

4.8.3 Relative Humidity. - The units shall be subjected to the humidity test of Specification MIL-T-5422. The units shall be tested to determine compliance with 3.16.3.

4.8.4 Orientation Test (See 3.16.4) - The units shall be inclined for a minimum of 5 minutes in each plane (forward, backward, left and right) to an angle of 20 $^{+3}_{-1}$ degrees. During inclination in each plane, the equipment shall meet specfied performance for the operation test of 4.18 to determine compliance with 3.27.

4.8.5 Sand and Dust (See 3.16.5). - The equipment shall be subjected to the sand and dust test of Specification MIL-T-5422. After the sand and dust tests, the equipment shall meet the requirements of 4.18 to determine compliance with 3.16.5.

4.8.6 Vibration. - The units shall be subjected to the vibration test of MIL-T-5422, for compliance with Curve IV of Table 2. Tests shall be performed before, during and after vibration to insure compliance with paragraphs 3.17.1 and 3.18.1 for compliance with 3.16.6.

4.8.7 Shock (See 3.16.7). -

4.8.7.1 Shock, Bench Handling (See 3.16.7.1). - The chassis and front panel assembly shall be removed from its enclosure, as for servicing, and placed in a suitable position for servicing on a solid 2-inch fir bench top. The test shall be performed as follows in a manner simulating shocks liable to occur during servicing:

a. Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axis for a total of 4 drops.

b. Repeat "a" with the assembly resting on other faces, until it has been dropped a total of four times on each face on which the assembly could be practically placed for servicing. Tests shall be performed before and after shock to insure compliance with paragraphs 3.17.1 and 3.18.1.

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4.8.7.2 Equipment Shock. - The unit(s) shall be subjected to the shock test of Specification MIL-T-5422. Tests shall be performed before and after shock to insure compliance with paragraphs 3.17.1 and 3.18.1 and for compliance with 3.16.7.2.

4.8.8 Mounting (Crash Safety). - The mounting shall be tested in accordance with MIL-T-5422 for compliance with 3.16.8.

4.8.9 Explosion Proof. - The units shall be subjected to the test of Specification MIL-T-5422 except as limited by paragraph 3.16.1 and for compliance with 3.16.9.

4.8.10 Fungus. - The unit(s) shall be subjected to 28 days of the fungus test specified in Method 508 of MIL-STD-810. For gasket or other non-hermetically sealed equipment, the parts and materials of the chassis shall be sprayed with the spore suspension. The interior of the case shall also be sprayed. The unit(s) shall be subjected to the test in this open condition and as specified for compliance with 3.16.10.

4.8.11 Salt Fog. - The unit(s) shall be exposed in its normal operating condition to salt fog in accordance with Method 509 of MIL-STD-810 for 48 hours. The unit(s) shall not be opened and shall not be required to be operated. The unit(s) shall meet the requirement of 3.16.11.

4.9 Electrical Tests, Transmitted (See 3.17). -

4.9.1 Power Output. - No modulation shall be applied. The power output shall be measured at frequencies of 30.00, 42.00, 52.95, 53.00, 65.00 and 75.95 MHz. The front panel switch shall be set to the low power setting and the test repeated to verify compliance with 3.17.1.

4.9.2 Distortion (See 3.17.2). -

4.9.2.1 Distortion, Narrow band. - The RF signal level to the receiver shall not be less than 50 microvolts. With transmitter operating under the appropriate standard test conditions, the total harmonic distortion measured at the audio output jack of the approved receiver with 1000 Hz applied, shall meet the requirements of 3.17.2.1. This measurement shall be made at one RF frequency.

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4.9.2.2 Distortion, Wideband. - With the transmitter deviated by a signal applied at the wideband terminal producing a 20 kHz deviation, the distortion shall be measured at the wideband output of a receiver which has the narrow band filters removed and replaced with dummy filters having input and output connected together internally. The transmitter output shall be attenuated to give an RF input to the receiver between 500 and 10,000 microvolts. Distortion measurements shall be made at 1,000, 5,000, 10,000 and 20,000 Hz to determine compliance with 3.17.2.2. This measurement shall be made at one RF frequency.

4.9.3 Side Tone. - With the AN/ARC-131 operating as a transmitter measure the side tone audio output noting its level compared to the audio level of the receiver under standard test conditions. Modulate the transmitter with standard audio input. Tests shall be conducted at 30.00, 52.95, and 75.95 MHz and supply voltage of 27.5 volts DC for compliance with 3.17.3.

4.9.4 Modulation Capability (See 3.17.4). - The transmitter shall be modulated at 500,1000, 2000, 3000, 10,000 and 20,000 Hz with 2.5 volts minimum applied to the "x -mode" input. At each modulation frequency the input shall be increased until the transmitter breaks lock. The maximum deviation reading before the transmitter breaks lock shall meet the requirements of 3.17.4. The test shall be conducted at transmitter dial frequencies of 35.00, 48.00, 57.00 and 70.00 MHz.

4.9.5 Modulator Sensitivity. - With the output of the transmitter connected to a frequency counter, the VFO control line shall be opened and ±1.0 VDC shall be applied to the VFO jack and the resultant frequencies shall be measured and recorded. The difference between the two resultant frequencies in kHz, divided by 2, is the modulator sensitivity in kHz/V. The test shall be conducted at transmitter frequencies of 30.00, 45.00, 52.00, and 53.00, 68.00, and 75.00 MHz for compliance with 3.17.5.

4.9.6 Resetability. - With the VFO control line open and the VFO jack terminated to ground thru a 50 OHM resistor, the receiver-transmitter shall be operated with no RF input, into a frequency counter thru a 50 OHM variable pad. The VFO frequency shall be measured after approaching the test frequency from both a clockwise and counterclockwise direction. This shall be done by setting the kHz and MHz knobs five detents below the test frequency where possible and then setting the kHz knob to frequency followed by setting the MHz knob to frequency. After measuring VFO frequency, the test frequency shall then be approached from a higher frequency in the same manner. The difference between the two VFO frequencies shall be considered the resetability. The test shall be conducted at test frequencies of 41.50, 51.50, 54.50, 64.50, and 74.50 MHz to meet the requirements of 3.17.6.

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4.9.7 Frequency Accuracy. - The output of the transmitter shall be attenuated to the proper level and fed into a frequency counter that has been properly calibrated. The transmitter output frequency shall be measured at dial frequencies of 75.00, 75.05, 75.10, 75.20, 75.30, 75.40, 75.50, 75.60, 75.70, 75.80, and 75.90 MHz. The deviation in kHz, from the frequency indicated by the dial shall be recorded to determine compliance with 3.17.7. This test shall also be conducted at -54°C and +65°C.

4.9.8 Maximum Signal-Plus-Noise to Noise Ratio. - For this test, the tone modulator shall be disabled. The transmitter dial frequency shall be 75.95 MHz. An approved receiver shall be tuned to the transmitter frequency and the S + N/N ratio shall be measured to determine compliance with 3.18.4. The level of the signal received by the approved receiver shall be 100 microvolts minimum. The requirements of 3.17.8 shall be met.

4.9.9 Audio Response (See 3.17.9). -

4.9.9.1 Narrow Band Audio Response. - With the transmitter operated to standard test conditions the transmitter shall be set up to 30.00 MHz and the deviation measured for the following modulating frequencies: 500, 1000, 5000, 10,000 and 20,000 Hz. The ratio expressed in db, of the deviation at various frequencies to that at 1000 Hz, shall be recorded and shall comply with 3.17.9.1. The test shall be repeated at 54.00 MHz.

4.9.9.2 Wideband Audio Response. - The transmitter shall be set-up to 30.00 MHz and the deviation measured for the following modulation frequencies: 50, 200, 1000, 5000, 10,000, and 20,000 Hz. The ratio expressed in db, of the deviation at the above frequencies to that at 1000 HZ, shall be recorded and shall comply with 3.17.9.2. The test shall be repeated at 54.00 MHz.

4.9.10 Audio Input Circuit Impedance. - The transmitter audio input circuit impedance shall be measured at 1000 CPS to determine compliance with 3.17.10.

4.9.11 Spurious Radiation. - No modulation shall be applied. The output of the transmitter shall be connected to an approved receiver covering the 30 to 76 MHz band through a 50 OHM variable pad to provide 70 millivolts across the receiver antenna terminals. With the transmitter on any one of the frequencies listed below, the receiver shall be tuned throughout its running range and points of quieting which exceed 10 DB shall be recorded to determine compliance with 3.17.11. The test shall be conducted at transmitter frequencies of 34.50, 46.00, 57.50, and 69.00 MHz. To distinguish between receiver and transmitter spurious radiation, quieting points shall be checked with a signal generator, tuned to frequency of quieting, substituted in place of the transmitter.

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the following power sources with characteristics as defined in MIL-[illegible] having limits

4.9.12 Deviation. - The deviation shall be measured at dial frequencies of 30.00, 52.95, 54.00, and 75.95 MHz to determine compliance with 3.17.12.

4.9.13 VFO Accuracy. - With the transmitter VFO control line shorted to ground with a 50 OHM resistor, the frequency of the transmitter oscillator shall be measured with a frequency counter at room temperature and recorded. The ambient temperature shall then be lowered to -40°C, held at this temperature for one hour, and the frequency again measured. The ambient temperature shall then be raised to +65°C, held at this temperature for one hour, and the frequency again measured. Measurements shall be made at dial frequencies of 30.50, 40.50, 51.50, 53.50, 63.50 and 74.50 MHz to determine compliance with 3.17.13.

4.9.14 Tone Deviation. - The transmitter shall be operated into a deviation meter through a suitable pad. The tone shall be transmitted when the squelch switch is set to "DIS", "CARR", and "TONE" positions and the deviation and frequency of the 150 Hz shall be measured to determine compliance with 3.17.14.

4.9.15 Catching Range. - A test unit consisting of an approved receiver tuner and crystal unit shall be used for this test. P-1002 (RF output from transmitter buffer) and P-1003 (IF output) shall be substituted for the equivalent connections in the transmitter. The transmitter shall be set to a dial frequency of 35.00 MHz and the test unit set on the compatible frequency. The receiver shall be tuned to the transmitter frequency and the tone modulation (1000 Hz at 0 DBM input) of the transmitter monitored. The transmitter shall then be detuned lower in frequency (minus catching range) than the original setting until the lock is broken, indicated by a loss of signal at the monitoring equipment. The transmitter shall then be tuned towards its original setting in 50 kHz increments until the signal at the monitoring receiver is regained. Record the difference between the original signal and the setting when the signal is regained. The procedure shall be repeated by detuning the transmitter higher in frequency, to determine the plus catching range. Repeat the above for dial frequencies at 48.00, 57.00 and 70.00 MHz. The requirements of 3.17.15 shall be met.

4.9.16 Continuous Transmit Operation. - The receiver-transmitter shall be exposed to a +65°C ambient temperature for at least three hours before testing. The receiver-transmitter shall be operated at +65°C ambient temperature in the high power transmit mode for three hours continuous transmission at 65.00 megacycles, into a dummy load representing a VSWR of 3 to 1. Compliance shall be determined by performance of the following tests and meeting the requirements of 3.17.16.

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XMTR Power Output	4.9.1 (run at 65.00 MHZ only)	3.17.1
Distortion	4.9.2 (run at 65.00 MHZ only)	3.17.2
Frequency Accuracy	4.9.7 (run at 65.00 MHZ only)	3.17.7
Deviation	4.9.12 (run at 65.00 MHZ only)	3.17.12

4.10 Receiver Performance Tests. (See 3.18). -

4.10.1 Sensitivity. - The RF input shall be 0.5 microvolt. The ratio of the signal-plus-noise plus distortion to noise plus distortion, expressed in DB, shall be measured with an appropriate distortion analyzer. Measurements shall be made at frequencies of 31.00, 41.00, 52.00, 53.00, 64.00 and 75.00 megacycles. At the temperature extremes (-54°C and +65°C), this test shall be performed with an RF input of 0.6 microvolts. Tests shall be made to assure compliance with 3.18.1.

4.10.2 Squelch Sensitivity (See 3.18.2). -

4.10.2.1 Tone Operation. - The squelch shall be set to the tone squelch position, and the signal generator modulated with a tone of 150 HZ, 3 KHZ deviation. With no RF signal applied to the receiver, the output shall be zero (0) and the retransmit contact on the squelch relay shall be open. Measure the RF input where the receiver unsquelches as the RF input is increased from 0.1 microvolt. Measure where the receiver again squelches as the RF input is reduced. Make measurements at 31.00, 41.00, 52.00, 53.00, 64.00 and 75.00 MHZ for compliance with 3.18.2.1.

4.10.2.2 Noise Operation. - The squelch control shall be set to the CARR position. With no RF signal applied to the receiver, the output shall be zero (0) and the retransmit contact on the squelch relay shall be open. Measure the RF input where the receiver unsquelches as the RF input is increased from 0.1 microvolt. Measure where the receiver again squelches as the RF input is reduced. Make measurements at 31.00, 41.00, 52.00, 53.00, 64.00 and 75.00 MHZ for compliance with 3.18.2.2.

4.10.2.3 Maximum Squelch Level. - Adjust squelch control for minimum sensitivity (fully clockwise) and measure RF input required to unsquelch the receiver. Make measurement at 31.00 and 64.00 MHZ and with 27.5 volt supply. Requirements of 3.18.2.3 shall be met.

4.10.2.4 Squelch Release. - With the squelch in carrier position apply a modulated signal of 1.0 volt or more and measure the time required for the receiver to reach standard audio output. The requirements of 3.18.2.4 shall be met.

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4.10.3 IF Selectivity. - A signal shall be injected into the input of the IF and the rectified signal at the output of the fourth stage of the IF shall be monitored. Adjust the input until this signal is at least 6 DB below saturation. With the signal level as a reference, adjust the input frequency and level to determine compliance with 3.18.3.

4.10.4 Maximum Signal-plus-noise to noise ratio. - The RF input shall be 100 microvolts. The ratio, expressed in DB, of the audio output with and without modulation shall be measured. The test shall be conducted at frequencies of 31.00, 41.00, 52.00, 64.00 and 75.00 MHZ for compliance with 3.18.4.

4.10.5 Desensitization. - Apply an RF signal to the receiver at a level to produce a 26 DB S + N/N ratio. An interfering signal which is ±0.5 MHZ removed from the desired RF signal shall be applied, and its level increased until the S + N/N ratio of the desired signal is reduced to 20 DB. The level of the interfering signal shall be recorded to determine compliance with 3.18.5. The test shall be conducted at frequencies of 30.00, 52.00, 53.00, and 75.00 MHZ.

4.10.6 Audio Output (See 3.18.6). -

4.10.6.1 Narrow Band Audio Output. - Under standard input conditions measure audio output to determine compliance with 3.18.6.1.

4.10.6.2 Wideband Audio Output. - Measure the audio output to determine compliance with 3.18.6.2.

4.10.7 Audio Response (See 3.18.7). -

4.10.7.1 Narrow Band Audio Response. - With the receiver operating under the standard test conditions the signal generator shall be modulated with an external audio oscillator and the deviation set to 8 KHZ at 1000 HZ modulated frequency. The audio output of the receiver shall be measured for frequencies of 150, 500, 1000, 2000, 3000, and 6000 HZ for compliance with 3.18.7.1.

4.10.7.2 Wideband Audio Response. - With the receiver operating under the standard test conditions the measurement shall be made at the wideband output. The test shall be performed in the same manner of that of 4.10.7.1 except measurements shall be made at the following audio frequencies: 50, 150, 300, 1000, 3000, 5000 and 12,000 HZ for compliance with 3.18.7.2.

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4.10.8 Limiting. - The RF input shall be 1.0 microvolt. The signal generator output shall then be increased to 100,000 microvolts and the audio power output continuously observed. The test shall be conducted at frequencies of 31.00, 41.00, 52.00, 64.00 and 75.00 megacycles, for compliance with 3.18.8.

4.10.9 Distortion (See 3.18.9). -

4.10.9.1 Narrow Band Distortion. - The distortion at the audio output shall be measured to determine compliance with 3.18.9.1. Measurements will be made at frequencies of 31.00 and 75.00 MHZ with RF input levels of 10 and 100,000 microvolts.

4.10.9.2 Wideband Distortion. - With the receiver operating under the standard test conditions the narrow band IF filters shall be removed and replaced with dummy filters having the input and output connected together internally. The RF input to the receiver shall be 200 microvolts. The distortion at the wideband audio output shall be measured at 50, 150, 300, 1000, 5000 and 12,000 HZ to determine compliance with 3.18.9.2.

4.10.10 Spurious Response. - The amount of quieting produced by an unmodulated RF carrier of 1.0 microvolt shall be used as a reference. The signal generator output shall be increased to 100,000 microvolts and the generator tuned throughout the range of 20 to 100 MHZ. Those frequencies which produce the quieting equal to or greater than the reference shall be recorded. The test shall be conducted at frequencies of 30.00 and 69.25 MC for compliance with 3.18.10.

4.10.11 IF Rejection. - The receiver shall be tuned to 30.00 MHZ. The amount of quieting produced by 1 microvolt shall be used as a reference. The signal generator shall then be tuned to 11.5 MHZ and its output raised until the amount of quieting is equal to the reference. The number of microvolts shall be recorded to determine compliance with 3.18.11. The test shall be repeated at frequencies of 52.00, 54.00 and 75.00 MHZ.

4.10.12 Intermodulation _____, - The receiver shall be tuned to 41.5 MHZ, and an unmodulated RF carrier of 1.0 microvolt shall be applied to the input of the receiver. The amount of quieting produced by this signal shall be used as a reference. Two generators, similar in all respects, tuned to 41.6 and 41.7 MHZ shall be coupled to the input of the receiver. The output of the generators shall be kept equal and simultaneously increased until the reference quieting is produced. The number of microvolts required to produce the reference shall be recorded to determine compliance with 3.18.12. The loading effect of the generators shall be considered when arriving at the number of microvolts required to produce the reference.

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4.10.13 Oscillator Radiation. - A field strength meter, or equivalent, shall be connected to the antenna terminals of the receiver. The receiver, shall be tuned to 30.0 MHZ and scanned with the field strength meter. The frequencies and amplitudes of detectable outputs greater than 75 microvolts shall be recorded to determine compliance with 3.18.13. This procedure shall be repeated for dial settings of 46.50, 53.20 and 75.60 MHZ.

4.10.14 VFO (See 3.18.14). -

4.10.14.1 Modulation Sensitivity. - The deviation of the local oscillator from the nominal frequency shall be measured when ±0.5 VDC is applied to the modulator input. With the control line open (no voltage applied to the modulator input), measure the VFO free running frequency. These measurements shall be taken at nominal VFO frequencies of 42.00 MHZ (53.50 MHZ dial frequency) 52.00 MHZ (63.50 MHZ dial frequency), and 63.00 MHZ (74.50 MHZ dial frequency) to determine compliance with 3.18.14.1.

4.10.14.2 Deviation Capability. - The deviation of the VFO from nominal frequency shall be measured when ±1.5 VDC is applied to the modulator input. With the control line open (no voltage applied to the modulator input), measure the VFO free running frequency. These measurements shall be taken at nominal VFO frequencies of 42.00 MHZ (53.50 MHZ dial frequency), 52.00 MHZ (63.50 MHZ dial frequency), and 63.00 MHZ (74.50 MHZ dial frequency) to determine compliance with 3.18.14.2.

4.10.14.3 Frequency Accuracy vs Temperature. - The automatic frequency control line of the VFO shall be grounded and the frequency of the VFO measured at room temperature. The ambient temperature shall then be lowered to -40°C, and maintained at that temperature for one hour, and then the VFO frequency shall be measured. The ambient temperature shall then be raised to +65°C, maintained for one hour, and the VFO frequency shall be measured. At each temperature step, the VFO frequency shall be measured at dial frequencies of 53.50, 63.50 and 74.50 MHZ. This test shall be conducted at input voltages of 24.0, 27.5 and 29.0 VDC. The requirements of 3.18.14.3 shall be met.

4.10.14.4 Repeatability. - With the VFO control line grounded and no input signal turn the KHZ and MHZ knobs at least 5 detents below the test frequency where possible. The KHZ knob shall be turned up to .50 MHZ followed by turning the MHZ knob up to 31 MHZ to give the test frequency of 31.50 MHZ. The frequency of the VFO shall be measured. Repeat the above, except the receiver shall be tuned to 31.50 MHZ from the direction of a higher frequency. The test shall be repeated for dial frequencies of 41.50 and 51.50 MHZ for compliance with 3.18.14.4.

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4.10.15 Maximum RF Input in Receiver. - Apply a 2 volt signal on frequency to the receiver for one minute. Following this, remove the signal and make measurements to determine that receiver performance has not been degraded. Perform test at 50.50 and 60.50 MHZ and with supply voltage at 27.5 VDC. The requirements of 3.18.15 shall be met.

4.10.16 Catching Range. - Disconnect all cables from the crystal switch assembly and connect to a crystal switch assembly in another receiver or to a test jig containing a crystal switch assembly. Tune the receiver under test to 30.90 MHZ, and the crystal switch to the appropriate corresponding frequency. Tune the dial of the receiver under test in 50 KHZ increments higher in frequency. After each movement of the dial, momentarily disable(short) the control line and observe whether or not the test-signal can be received. Record the last dial reading at which the receiver is capable of receiving 30.90 MHZ then tune the radio set and signal generator to 31.90 MHZ, and then tune the radio set lower in 50 KHZ increments until the signal is lost. Record the last frequency at which the signal is heard. The plus and minus catching ranges and their sum for each pair of frequencies shall meet the requirement of 3.18.16. This test shall be repeated at each of the following pairs of frequencies:

(30.90, 31.90), (31.95, 32.95), (52.90, 33.90), (33.95, 34.95),
(34.90, 35.90), (35.95, 36.95), (36.90, 37.90), (37.95, 38.95),
(38.90, 39.90), (39.95, 40.95), (40.90, 41,90), (41.95, 42.95),
(42.90, 43.90), (43.95, 44.95), (44.90, 45.90), (45.95, 46.95),
(46.90, 47.90), (47.95, 48.95), (48.90, 49.90), (49.95, 50.95),
(50.90, 51,90), and (51.95, 52.95).

4.10.17 Residual Phase Detection Output. - The control loop shall be open (error signal lead connected). Connect a DC VTVM so that the output of the phase detector only is measured. With the receiver dial set to 41.45 MHZ, and with a 250 millivolts RF signal applied into the crystal reference system high pass filter, measure the residual DC output voltage while varying the RF signal frequency from 52.55 to 53.35 MHZ. Disregard the output reading at 52.95 MHZ. The requirements of 3.18.17 shall be met.

4.10.18 Blocking. - Apply two RF signals with squelch in disable position, tuned to the center frequency of the receiver, to the input of the receiver. Adjust one RF input for 1 microvolt 8 KHZ deviation at 1000 HZ and the second RF for one volt unmodulated. Measure the time required for the audio output to recover following the removal of the one volt signal. Make tests at carrier frequencies of 32.15, 40.00, 51.65, 59.95 and 64.85 MHZ for compliance with 3.18.18.

4.11 Homing (See 3.19). -

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4.11.1 Homing Sensitivity. - All homing measurements are to be made in the following test set up:

The signal generator output shall be connected to a fixed 6.0 DB 50 ohm pad. The output of the fixed 6 DB pad shall be connected to the input of two variable attenuators using a Tee Connector UG-274/U. The variable 50 ohm attenuators shall be set to 6.0 DB. All signal levels for homing tests are the levels at the input of the fixed 6 DB pad. Vary the RF input above and below squelch operating the flags. Both vertical and horizontal flags of the ID-48/ARN should appear when the receiver is squelched, and disappear when the receiver is unsquelched. Measure at 52.00 MHZ for compliance with 3.19.1.

4.11.2 Visual Homing Indication. - Under standard test conditions with standard test signal set for receive operation, place the receiver in "HOME" and "CARR" position. From a common generator apply signal to the homing antenna input. Adjust level of the input so that (1) right is stronger than left, (2) left is stronger than right, (3) equal amplitudes are applied to both inputs, and (4) signal strength is varied over the levels specified. Observe the vertical needle of ID-48. Measure at carrier frequencies of 31.00, 52.00 and 64.00 MHZ for compliance with 3.19.2.

4.11.3 Isolation of Communication Antenna Input During Homing. - Set up equipment as in paragraph 4.11.1. Reduce one of the variable attenuators to 3 DB for a 3 DB homing differential. Apply 0.5 millivolts at the input of the variable attenuators and not the deflection of the vertical needle. While maintaining the input to the homing antenna leads, apply a 5.0 millivolt signal to the Communications Antenna lead. Note the change in the vertical needle deflection for compliance with 3.19.3.

4.12 Frequency Selection. - Using a stopwatch, monitoring the system visually and by means of the channel changing tone, measure the time required for the RT unit to change channels. Make measurements at the following frequencies and for supply voltages of 24, 27.5 and 29 volts. (See 3.20).

a. 74.00 to 52.05 MHZ

b. 51.00 to 74.05 MHZ

4.13 Input Voltage (See 3.21). -

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4.13.1 Input Voltage Range Test. - The following tests shall be conducted at input voltages of 24.0 and 29.00 VDC for compliance with paragraph 3.21.1.

a. Power output 4.9.1

b. Distortion 4.9.2.1

c. Audio output 4.10.6

d. Sensitivity 4.10.1

e. Distortion 4.10.9

4.13.2 Transient Protection (See 3.21.2). - Connect a diode between the power source and the Radio Set to isolate the power source. A 25,000 uf capacitor shall be charged to 80 volts and then connected across the power input terminals of the radio set while operating. This will produce a triangular pulse of 80 V peak amplitude at the input of the radio set. This test shall be performed in both receive and transmit modes.

4.14 Retransmission. - Connect the two AN/ARC-131 systems for retransmit. Set the function switch of each to RETRANSMIT position and each squelch switch to the same position ("Carrier" or "Tone" position). Verify that an RF input of 0.8 microvolts deviated ± 8 KHZ at 1000 HZ applied to the communication antenna terminals of either system will cause the non-receive system to have power output containing 1000 HZ modulation. Tests shall be conducted on 40.5 and 60.5 MHZ/Sec. The requirements of 3.22 shall be met.

4.15 Radio Interference. - The equipment shall be tested in accordance with Specification MIL-I-6181 for compliance with paragraph 3.23.

4.16 I-Mode. - Unless otherwise specified, the equipment shall be tested by the Government for compliance with 3.24.

4.17 Inspection for Interchangeability. - The mechanical dimensions and electrical performance required by the mechanical gage drawings and electrical test fixture drawing listed below shall be gaged or measured to determine conformance to the physical and functional interchangeability requirements of 3.25. When a listed dimension or performance requirement is not within specified or design limits, it shall be considered a major defect. The following dimensions or performance shall be gaged or measured:

a. Mechanical (Dimension). - The gage points covered by the drawings listed on the applicable gage list as listed in the contract for this order.

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b. Electrical Performance (Modules). - Prior to assembly into the equipment, plug-in modules shall be tested to determine compliance with the performance requirements specified in the applicable test procedure and gage lists as listed in the contract for this order.

4.18 Operation Inspection Test. - Each unit on order shall be combined with corresponding components and associated equipment to form an operational unit. (Substitution of simulated associated equipment is permissible when operational conditions are accurately duplicated). Power shall be supplied at low line voltage (24 volts DC). Satisfactory transmission and reception quality shall indicate proper operation of the equipment. This test shall include a communication check with equipment operating in the homing mode. With the system operating normally, check all positions of the performance indicating meter for compliance with 3.27.

4.19 Visual and Mechanical Inspection. - Unit(s) shall be examined for the defects listed in MIL-STD-252. Prior to assembly into the unit(s) all printed circuit boards shall be examined in accordance with MIL-P-55110, and all modules and sections shall be examined in accordance with MIL-STD-252. These inspections shall be performed to determine compliance with 3.29.

4.20 Quality Conformance Inspection of Preparation for Delivery. - Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

4.21 Rough Handling Test Preparation for Delivery. - When rough handling test in accordance with Specification MIL-P-116 is required by the contract (See 6.2), the following functional test shall be conducted to determine freedom from operational malfunction caused by rough handling: 4.18 Operational Test.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging. - Preservation and packaging of Radio Set AN/ARC-131 and Units of shall be level A or C as specified (See 6.2).

5.1.1 Level A. -

5.1.1.1 Cleaning. - Each radio set and units of shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. - Each radio set and units of shall be dried in accordance with the applicable procedure of MIL-P-116.

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5.1.1.3 Preservation Application. - None required.

5.1.1.4 Unit Packaging. - Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical Literature. - Each technical literature shall be packaged method IC-1.

5.1.1.4.2 Radio Set AN/ARC-131. - Each set shall be packaged individually in accordance with method III of MIL-P-116. Install the receiver-transmitter on the mounting and secure with the applicable devices. Individually cushion the receiver-transmitter installed on the mounting and the radio set control on all surfaces with cells or pads or both fabricated of fiberboard conforming to PPP-F-320, type CF, class weather-resistant, variety SW, grade W5c, designed to absorb the shock of impact in handling and transit. Place the cushioned items within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Fill all voids with cells or pads conforming to the specification of the cushioning material or dummy boxes conforming to the specification of the outer container to prevent movement during handling and transit. As required by the contract or order, place the technical literature, packaged as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.1.4.3 Receiver-Transmitter RT-823( )/ARC-131, Radio Set Control C-7088( )/ARC-131 or Mounting MT-3664( )/ARC-131. - Each receiver-transmitter, radio set control or mounting shall be packaged individually in accordance with method III of MIL-P-116 in the manner described in 5.1.1.4.2.

5.1.2 Level C. - Each radio set or units of shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. - Packing of Radio Set AN/ARC-131 and Units Of shall be level A, B or C as specified (See 6.2).

5.2.1 Level A. -

5.2.1.1 A quantity of radio sets or units of, packaged as specified in 5.1 and each bearing the same stock number, shall be packed together within a close-fitting, fiberboard box conforming to PPP-B-636, type CF, class weather-resistant, special requirements use; PPP-B-640, class 2, style E or MIL-B-43291. The gross weight of boxes conforming to PPP-B-640 shall not exceed 200 pounds. Close the box in accordance with the appendix of the applicable box specification. To facilitate palletization, fiberboard boxes shall be uniform in size and contain equal quantities to the greatest extent practicable.

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5.2.1.2 Palletized Load. - A quantity of radio sets or units of, each bearing the same stock number and packed as specified in 5.2.1.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s. The load shall be "bonded" to the pallet by strapping.

5.2.1.3 Less than Palletized Load. - When quantities per destination are less than a pallet load, containers specified in 5.2.1.1 shall be reinforced by pressure-sensitive filament tape or nonmetallic strapping conforming to PPP-T-97, type IV, and PPP-S-760, type II, respectively. Selection of the material and application shall be in accordance with the appendix of the applicable box specification. No further packing shall be required.

5.2.2 Level B. - Each radio set and units of, packaged as specified in 5.1, shall be packed as specified in 5.2.1 (See 6.2 and 6.5).

5.2.3 Level C. - Each radio set and units of, packaged as specified in 5.1, shall be packed as specified in 5.2.1 except that the pallets, fiberboard boxes and fiberboard caps shall be of the type, size and kind commonly used for the purpose and shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation. Reinforcing of the boxes, as specified in 5.2.1.3, will not be required. Packing shall be accomplished in a manner that will afford adequate protection to the package and its contents during shipment from the supply source to the first receiving activity.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended Use. - Radio Set AN/ARC-131, when installed in aircraft provides two-way radio communication by means of FM radio telephone (F3) on any frequency from 30.00 to 75.95 MHZ, homing on any radio signal of sufficient signal strength within its frequency range, and retransmission in an automatic, unattended radio repeater type of operation.

6.2 Ordering Data. - Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

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b. Level of packaging and level of packing required for shipment. (Level A, B or Level C) as specified in contract.

c. The specific packaging and packing which are applicable to the particular procurement.

d. Preproduction Inspection.

(1) Three sample units of each item cited in section 1 are generally required so that lengthy environmental and reliability tests can be completed on specified sample units while complete performance measurements can be made on the one sample unit. (See 3.3)

(2) Preproduction pack(s) as follows:

(a) Makeup of pack(s).

(b) Number of each kind of pack to be submitted.

(c) Inspection to be performed thereon, including rough handling.

(d) Marking and shipping of samples.

(e) Place of final inspection.

(f) Technical literature required. (See 3.28)

(g) Quantity of tools and spare parts required.

(h) Submission of the statement of treatment referenced in 3.15.4 within 30 days after award of contract. This statement should be submitted with the tests to be performed during the contract, to the contracting officer.

6.3 Nomenclature. - The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AN/ARC-131W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.4 Location of Operational Inspection. - It is desirable that the operational inspection 4.13 be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be given the operational inspection immediately prior to packaging.

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6.5 Definition of Level B. - Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments; covered storage for a period of three years or less; and distribution within the continental limits of the United States.

6.6 Verification Inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in the specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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